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Laporan Akhir

Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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1.1.4 Tabel Link Antar Router

Tabel 3: Konfigurasi Link Antar Perangkat

Link	Network	Perangkat	Interface	IP Address
Cisco ↔ Huawei Link 1	10.10.10.0/30	Cisco Router	Gi0/3	10.10.10.1/30
Cisco ↔ Huawei Link 1	10.10.10.0/30	Huawei Router	Ethernet1/0/0	10.10.10.2/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Cisco Router	Gi0/0	10.10.11.1/30
Cisco ↔ Huawei Link 2	10.10.11.0/30	Huawei Router	Ethernet1/0/4	10.10.11.2/30
Cisco ↔ MikroTik	10.10.30.0/30	Cisco Router	Gi0/1	10.10.30.2/30
Cisco ↔ MikroTik	10.10.30.0/30	MikroTik	ether2	10.10.30.1/30
MikroTik ↔ Huawei	10.10.20.0/30	MikroTik	ether1	10.10.20.2/30
MikroTik ↔ Huawei	10.10.20.0/30	Huawei Router	Ethernet1/0/3	10.10.20.1/30

Praktikum 1 Konfigurasi Cisco Switch

1. Masuk ke Terminal Cisco Switchh

```
1 Switch> enable
2 Switch# configure terminal
3 Switch(config)#
4
```

2. Membuat vlan

```
Switch(config)#vlan 10
Switch(config-vlan)#name finance
Switch(config-vlan)#exit
Switch(config)#
*Jun  5 06:19:35.583: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerpr
isek9-m' passed code signing verification
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 40
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name Marketingg
Switch(config-vlan)#exit
```

Gambar 2: membuat vlan

3. show vlan brief

```
LAN Name:         Status      Ports
-----
Vl001 default      active     Gi0/0, Gi0/1, Gi0/2, Gi0/3
Vl002 finance      active     Gi1/0, Gi1/1, Gi1/2, Gi1/3
Vl003 HR            active
Vl004 IT            active
Vl005 Marketingg    active
Vl002 fddi-default  act/unsup
Vl003 token-ring-default  act/unsup
Vl004 fddinet-default  act/unsup
```

Gambar 3: vlan sudah dibuat

4. konfigurasi Access port untuk setiap vlan

```

Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/1
Switch(config-if)#description PC-HR-VLAN20
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/2
Switch(config-if)#description PC-FINANCE-VLAN10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi1/1
Switch(config-if)#description PC-IT-VLAN30
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi0/3
Switch(config-if)#description PC-MARKETING-VLAN40
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#show vlan brief
Switch#
VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi1/0, Gi1/2, Gi1/3
10   finance                 active    Gi0/2
20   HR                      active    Gi0/1
30   IT                      active    Gi1/1
40   Marketingg              active    Gi0/3
1002 fddi-default           act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#

```

Gambar 4: konfigurasi access point

5. konfigurasi Trunk ke cisco router

```

Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#description TRUNK-TO-CISCO-ROUTER
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#show interfaces trunk
Switch#
Port      Mode          Encapsulation  Status        Native vlan
Gi0/0     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     none
Switch#

```

Gambar 5: konfigurasi dan show interface

```

Switch#write memory
Building configuration...
Compressed configuration from 3364 bytes to 1653 bytes[OK]
Switch#
Jun  5 06:29:49.844: %GRUB-5-CONFIG-WRITING: GRUB configuration is being updated on disk. Please wait...
Jun  5 06:29:50.555: %GRUB-5-CONFIG-WRITTEN: GRUB configuration was written to disk successfully.
Switch#

```

Gambar 6: write memory

Praktikum 2 Konfigurasi Cisco Router

1. Masuk ke cisco Router

2. Konfigurasi Interface Trunk ke Switch
3. Konfigurasi Subinterface VLAN 10
4. Konfigurasi Subinterface VLAN 20
5. Konfigurasi Subinterface VLAN 30
6. Konfigurasi Subinterface VLAN 40
7. Konfigurasi DHCP-server untuk VLAN 10
8. Konfigurasi DHCP-server untuk VLAN 20

```

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/2
Router(config-if)#description TRUNK-T0-CISCO-SWITCH
Router(config-if)#no ip address
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gig
*Jun  5 06:31:15.579: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
*Jun  5 06:31:16.579: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, change
Router(config)#interface gig0/2.10
Router(config-subif)#description GATEWAY-VLAN10-FINANCE
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.30
Router(config-subif)#description GATEWAY-VLAN30-IT
Router(config-subif)#encapsulation dot1q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.20
Router(config-subif)#description GATEWAY-VLAN20-HR
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#interface gig0/2.40
Router(config-subif)#description GATEWAY-VLAN40-MARKETING
Router(config-subif)#encapsulation dot1q 40
Router(config-subif)#ip address 192.168.40.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool VLAN10-FINANCE
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.10
Router(config)#ip dhcp pool VLAN20-HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#

```

Gambar 7: langkah 1-8

9. Konfigurasi IP static pada VPCS untuk Vlsn 30 dan 40

```

Press '?' to get help.

VPCS> ip 192.168.30.10/24 192.168.30.1
Checking for duplicate address...
PC1 : 192.168.30.10 255.255.255.0 gateway 192.168.30.1

VPCS>

```

Gambar 8: vlan 30

```

Press '?' to get help.

VPCS> ip 192.168.40.10/24 192.168.40.1
Checking for duplicate address...
PC1 : 192.168.40.10 255.255.255.0 gateway 192.168.40.1

VPCS>

```

Gambar 9: vlan 40

```

Press '?' to get help.

VPCS>
VPCS> ip dhcp
DDORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS>

```

Gambar 10: ip dhcp

10. verifikasi cisco router

```

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset   administratively down  down
GigabitEthernet0/1    unassigned      YES unset   administratively down  down
GigabitEthernet0/2    unassigned      YES unset   up          up
GigabitEthernet0/2.10 192.168.10.1    YES manual  up          up
GigabitEthernet0/2.20 192.168.20.1    YES manual  up          up
GigabitEthernet0/2.30 192.168.30.1    YES manual  up          up
GigabitEthernet0/2.40 192.168.40.1    YES manual  up          up
GigabitEthernet0/3    unassigned      YES unset   administratively down  down
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 0
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.10.1   192.168.10.1 - 192.168.10.254  0

Pool VLAN20-HR :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 1
  Pending event : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.20.12  192.168.20.1 - 192.168.20.254  1
Router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
192.168.20.11   0100.5079.6668.5b    Jun 06 2026 06:42 AM  Automatic
Router#

```

Gambar 11: show ip interface brief, show ip dhcp

11. pengujian dari VPCS

```

VPCS> ip dhcp
DDORA IP 192.168.10.11/24 GW 192.168.10.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.10.11/24
GATEWAY    : 192.168.10.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.10.1
DHCP LEASE  : 86382, 86400/43200/75600
MAC        : 00:50:79:66:68:5c
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU        : 1500

VPCS>

```

Gambar 12: ip dhcp dan show ip vlan 10

```
VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.062 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.630 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.431 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.563 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.261 ms

VPCS> █
```

Gambar 13: ping dari vlan 10

```
VPCS> ip dhcp
DORA IP 192.168.20.11/24 GW 192.168.20.1

VPCS> show up
Invalid arguments

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.20.11/24
GATEWAY    : 192.168.20.1
DNS        : 8.8.8.8
DHCP SERVER : 192.168.20.1
DHCP LEASE : 86398, 86400/43200/75600
MAC        : 00:50:79:66:68:5b
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=2.917 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=3.529 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=3.835 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=3.264 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=4.257 ms

VPCS> █
```

Gambar 14: ip dhcp, show ip dan ping dari vlan 20

```
VPCS> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=255 time=4.800 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=255 time=3.215 ms
84 bytes from 192.168.30.1 icmp_seq=3 ttl=255 time=8.025 ms
84 bytes from 192.168.30.1 icmp_seq=4 ttl=255 time=5.262 ms
84 bytes from 192.168.30.1 icmp_seq=5 ttl=255 time=5.369 ms

VPCS> █
```

Gambar 15: ping dari vlan 30

```
VPCS> ip dhcp
DDD
Can't find dhcp server

VPCS> ping 192.168.40.1

84 bytes from 192.168.40.1 icmp_seq=1 ttl=255 time=4.355 ms
84 bytes from 192.168.40.1 icmp_seq=2 ttl=255 time=11.174 ms
84 bytes from 192.168.40.1 icmp_seq=3 ttl=255 time=4.965 ms
84 bytes from 192.168.40.1 icmp_seq=4 ttl=255 time=10.370 ms

█
```

Gambar 16: ip dhcp dan show ip vlan 40

```

VPCS> ping 192.168.10.1

84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=3.062 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=3.630 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=3.431 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=3.563 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=3.261 ms

VPCS> ping 192.168.20.11

84 bytes from 192.168.20.11 icmp_seq=1 ttl=63 time=6.553 ms
84 bytes from 192.168.20.11 icmp_seq=2 ttl=63 time=5.699 ms
84 bytes from 192.168.20.11 icmp_seq=3 ttl=63 time=5.704 ms
84 bytes from 192.168.20.11 icmp_seq=4 ttl=63 time=7.529 ms
84 bytes from 192.168.20.11 icmp_seq=5 ttl=63 time=4.790 ms
p
VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=63 time=5.839 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=63 time=5.358 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=63 time=4.744 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=63 time=7.320 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=63 time=4.467 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=7.026 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=5.814 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.744 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=63 time=4.584 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=63 time=5.839 ms

VPCS>

```

Gambar 17: uji ping antar valn dari vlan 10

```

router#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type
                Hardware address/
                User name
192.168.10.11    0100.5079.6668.5c    Jun 06 2026 06:44 AM    Automatic
192.168.20.11    0100.5079.6668.5b    Jun 06 2026 06:45 AM    Automatic
router#

```

Gambar 18: ip dhcp binding

Praktikum 3 Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi Lan Huawei

```

<HUAWEI>system-view
Enter system view, return user view with return command.
[~HUAWEI]interface Ethernet1/0/2
[~HUAWEI-Ethernet1/0/2]description LAN-HUAWEI
[*HUAWEI-Ethernet1/0/2]ip address 192.168.50.1 255.255.255.0
[*HUAWEI-Ethernet1/0/2]undo shutdown
[*HUAWEI-Ethernet1/0/2]quit
[*HUAWEI]commit
[~HUAWEI]

```

Gambar 19: confi lan huawei

2. Konfigurasi IP VPC Lan huawei

```
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:5d
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500
```

Gambar 20: show ip

```
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=13.408 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=3.324 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=3.915 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=3.182 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=3.026 ms
```

Gambar 21: cping gateway 50.1

3. Konfigurasi IP link cisco antar huawei link 1

```
Router(config)#interface gi0/3
Router(config-if)#description LINK1-T0-HUAWEI
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown exit
% Invalid input detected at '^' marker.
Router(config-if)#no shutdown
Router(config-if)#exit
```

Gambar 22: cisco ruter

```
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/0
[~HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ip address 10.10.10.2 255.255.255.252
[*HUAWEI-Ethernet1/0/0]undo shutdown
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 23: huawei ruter

4. konfigurasi ip link cisco antar huawei link 2

```
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip address 10.10.11.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Gambar 24: cisco ruter

```
[~HUAWEI]interface Ethernet1/0/4
[~HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ip address 10.10.11.2 255.255.255.252
[*HUAWEI-Ethernet1/0/4]undo shutdown
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]commit
```

Gambar 25: huawei ruter

5. Konfigurasi IP link Huawei antar mikrotik

```
Error: Unrecognized command found at '^' position.
[~HUAWEI]interface Ethernet1/0/3
[~HUAWEI-Ethernet1/0/3]description LINK-TO-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ip address 10.10.20.1 255.255.255.252
[*HUAWEI-Ethernet1/0/3]undo shutdown
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit
[~HUAWEI]
```

Gambar 26: huawei ruter

6. Konfigurasi IP Link cisco dengan mikrotik

```
Router(config)#interface gi0/1
Router(config-if)#description LINK-TO-MIKROTIK
Router(config-if)#ip address 10.10.30.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
```

Gambar 27: cisco ruter

```
admin@mikrotik] >> /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
```

Gambar 28: mikrotik

7. verivikasi interface

```
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.10.11.1 YES manual up
GigabitEthernet0/1 10.10.30.1 YES manual up
GigabitEthernet0/2 unassigned YES unset up
GigabitEthernet0/2.10 192.168.10.1 YES manual up
GigabitEthernet0/2.20 192.168.20.1 YES manual up
GigabitEthernet0/2.30 192.168.30.1 YES manual up
GigabitEthernet0/2.40 192.168.40.1 YES manual up
GigabitEthernet0/3 10.10.10.1 YES manual up
Router#
```

Gambar 29: cisco ruter

```
[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface IP Address/Mask Physical Protocol VPN
Ethernet1/0/0 10.10.10.2/30 up up --
Ethernet1/0/0.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/1 unassigned up down --
Ethernet1/0/1.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/2 192.168.50.1/24 up up --
Ethernet1/0/2.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/3 10.10.20.1/30 up up --
Ethernet1/0/3.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/4 10.10.11.2/30 up up --
Ethernet1/0/4.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/5 unassigned up down --
Ethernet1/0/5.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/6 unassigned up down --
Ethernet1/0/6.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/7 unassigned up down --
Ethernet1/0/7.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/8 unassigned up down --
Ethernet1/0/8.4094 128.1.138.137/16 up up l3vpn
Ethernet1/0/9 unassigned up down --
Ethernet1/0/9.4094 128.1.138.137/16 up up l3vpn
GigabitEthernet0/0/0 unassigned up down --
LoopBack2147483647 128.1.138.137/16 up up(s) l3vpn
NULL0 unassigned up up(s) --
[~HUAWEI]
```

Gambar 30: Huawei ruter

```
[admin@MikroTik] >> /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;;; TO-HUAWEI
10.10.20.2/30 10.10.20.0 ether1
1 ;;; TO-CISCO
10.10.30.2/30 10.10.30.0 ether2
```

Gambar 31: mikrotik

8. verifikasi ping antar router

```
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/4 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/8/28 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

Gambar 32: cisco ruter

```
[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/2/3 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=4 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=4 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 2/3/4 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms
```

Gambar 33: huawei

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                            56 255 4ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 1ms
4 10.10.20.1                            56 255 1ms
5 10.10.20.1                            56 255 1ms
6 10.10.20.1                            56 255 2ms
sent=7 received=7 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                            56 255 2ms
1 10.10.30.1                            56 255 3ms
2 10.10.30.1                            56 255 1ms
sent=3 received=3 packet-loss=0% min-rtt=1ms avg-rtt=2ms max-rtt=3ms
```

Gambar 34: mikrotik

Praktikum 4 Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konfigurasi ospf pada cisco router

```

Router(config)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#network 10.10.10.0 0.0.0.3 area 0
Router(config-router)#network 10.10.11.0 0.0.0.3 area 0
Router(config-router)#network 10.10.30.0 0.0.0.3 area 0
Router(config-router)#network 192.168.10.0 0.0.0.255 area 0
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.30.0 0.0.0.255 area 0
Router(config-router)#network 192.168.40.0 0.0.0.255 area 0
Router(config-router)#exit
Router(config)#interface gi0/3
Router(config-if)#description LINK-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/0
Router(config-if)#description LINK2-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/3
Router(config-if)#description LINK1-T0-HUAWEI
Router(config-if)#ip ospf cost 10
Router(config-if)#exit
Router(config)#interface gi0/1
Router(config-if)#description LINK-T0-MIKROTIK
Router(config-if)#ip ospf cost 100
Router(config-if)#exit

```

Gambar 35: config ospf

```

Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 110)

```

Gambar 36: verifikasi ospf

2. Konfigurasi OSPF pada Huawei router

```

[~HUAWEI]system-view
Error: Unrecognized command found at '^' position.
[~HUAWEI]ospf 1 router-id 2.2.2.2
[*HUAWEI-ospf-1]area 0.0.0.0
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.10.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.11.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 10.10.20.0 0.0.0.3
[*HUAWEI-ospf-1-area-0.0.0.0]network 192.168.50.0 0.0.0.255
[*HUAWEI-ospf-1-area-0.0.0.0]quit
[*HUAWEI-ospf-1]quit
[*HUAWEI]interface Ethernet1/0/0
[*HUAWEI-Ethernet1/0/0]description LINK1-T0-CISCO
[*HUAWEI-Ethernet1/0/0]ospf cost 10
[*HUAWEI-Ethernet1/0/0]quit
[*HUAWEI]interface Ethernet1/0/4
[*HUAWEI-Ethernet1/0/4]description LINK2-T0-CISCO
[*HUAWEI-Ethernet1/0/4]ospf cost 10
[*HUAWEI-Ethernet1/0/4]quit
[*HUAWEI]interface Ethernet1/0/3
[*HUAWEI-Ethernet1/0/3]description LINK-T0-MIKROTIK
[*HUAWEI-Ethernet1/0/3]ospf cost 100
[*HUAWEI-Ethernet1/0/3]quit
[*HUAWEI]commit

```

Gambar 37: config pada huawei

```

[HUAWEI]display ospf peer

M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
        Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2        MTU: 1500
  Dead timer due in 36 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m28s
  Neighbor Up Time : 2026-06-05 07:15:57
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1            BDR: 10.10.11.2        MTU: 1500
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m28s
  Neighbor Up Time : 2026-06-05 07:15:57
  Authentication Sequence: [ 0 ]

```

Gambar 38: verif ospf huawei

```

192.168.10.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
                  OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.20.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
                  OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.30.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
                  OSPF 10 11 D 10.10.10.1 Ethernet1/0/0
192.168.40.0/24 OSPF 10 11 D 10.10.11.1 Ethernet1/0/4
                  OSPF 10 11 D 10.10.10.1 Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4 Routes : 4

Destination/Mask Proto Pre Cost Flags NextHop Interface
-----
10.10.10.0/30 OSPF 10 10 10.10.10.2 Ethernet1/0/0
10.10.11.0/30 OSPF 10 10 10.10.11.2 Ethernet1/0/4
10.10.20.0/30 OSPF 10 100 10.10.20.1 Ethernet1/0/3
192.168.50.0/24 OSPF 10 1 192.168.50.1 Ethernet1/0/2

-HUAMEI#display ospf routing

OSPF Process 1 with Router ID 2.2.2.2
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
-----
10.10.10.0/30 10 Direct 10.10.10.2 2.2.2.2 0.0.0.0
10.10.11.0/30 10 Direct 10.10.11.2 2.2.2.2 0.0.0.0
10.10.20.0/30 100 Direct 10.10.20.1 2.2.2.2 0.0.0.0
10.10.30.0/30 110 Stub 10.10.11.1 1.1.1.1 0.0.0.0
10.10.30.0/30 110 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.10.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.20.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.30.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.11.1 1.1.1.1 0.0.0.0
192.168.40.0/24 11 Stub 10.10.10.1 1.1.1.1 0.0.0.0
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables

Routing for Network
Destination Cost Type NextHop AdvRouter Area
-----
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0

```

Gambar 39: display ospf routing huawei

3. Konfigurasi OSPF pada mikrotik

[illegible]

Gambar 40: network mikrotik

[illegible]

Gambar 41: ip route ospf

4. Verifikasi OSPF neighbor

```
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/DR 00:00:30 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:32 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:29 10.10.10.2 GigabitEthernet0/3
Router#
```

Gambar 42: show ip neighbor

```
192.168.50.0/24 1 Direct 192.168.50.1 2.2.2.2 0.0.0.0
Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

OSPF Process 65534 with Router ID 128.1.138.137
Routing Tables
Routing for Network
Destination Cost Type NextHop AdvRouter Area
128.1.138.137/32 0 Direct 128.1.138.137 128.1.138.137 0.0.0.0
Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
~HUAWEI~display ospf peer
M) Indicates MADJ neighbor

OSPF Process 1 with Router ID 2.2.2.2
Neighbors
Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.10.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.10.1 BDR: 10.10.10.2 MTU: 1500
Dead timer due in 30 sec
Retrans timer interval: 5
Neighbor is up for 00h05m40s
Neighbor Up Time : 2026-06-05 07:15:57
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3 Address: 10.10.20.2
State: Full Mode:Nbr is Master Priority: 1
DR: 10.10.20.1 BDR: 10.10.20.2 MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h02m02s
Neighbor Up Time : 2026-06-05 07:19:35
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1 Address: 10.10.11.1
State: Full Mode:Nbr is Slave Priority: 1
DR: 10.10.11.1 BDR: 10.10.11.2 MTU: 1500
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00h05m40s
Neighbor Up Time : 2026-06-05 07:15:57
Authentication Sequence: [ 0 ]
```

Gambar 43: display peer huawei

```
Router#show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/DR 00:00:30 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/DR 00:00:32 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/DR 00:00:29 10.10.10.2 GigabitEthernet0/3
Router#
```

Gambar 44: neighbor ospf mikrotik

5. Verifikasi Routing table

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
Known via "ospf 1", distance 110, metric 11, type intra area
Last update from 10.10.11.2 on GigabitEthernet0/0, 00:06:50 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:06:50 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:06:50 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#
```

Gambar 45: show ip route 192.168.50.0

Gambar 46: show ip route ospf cisco

Gambar 47: ip route mikrotik

Gambar 48: dari valn 10

Gambar 49: dari vlan 20

Gambar 50: dari vlan 30

Gambar 51: dari vlan 40

Gambar 52: dari lan huawei

1. Kondisi normal: Dua link cisco-huawei aktif

Gambar 53: show ip 50.0

Gambar 54: trace dari vc vlan

[illegible]

15

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.40.1    3.219 ms  3.925 ms  3.348 ms
 2  10.10.11.2     6.402 ms  4.779 ms  4.677 ms
 3  *192.168.50.10  9.711 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 56: trace 50.10

3. Kondisi dua link cisco huawei mati

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:37 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:37 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router#

```

Gambar 57: show ip route 50.0

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.40.1    3.506 ms  2.752 ms  3.190 ms
 2  10.10.30.2     3.805 ms  2.907 ms  4.106 ms
 3  10.10.20.1     7.848 ms  5.701 ms  4.703 ms
 4  *192.168.50.10  6.901 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 58: trace 50.10

4. mengaktifkan kembali dua link cisco huawei

```

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/3
Router(config-if)#no shutdown
Router(config-if)#
*Jun 5 07:44:42.587: %LINK-3-UPDOWN: Interface GigabitEthernet0/3, changed state to up
*Jun 5 07:44:43.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/3, changed state to up
Router(config-if)#exit
Router(config)#interface gig0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Jun 5 07:45:03.856: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 5 07:45:04.857: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to u
Router(config)#
*Jun 5 07:45:09.447: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router(config)#
% Invalid input detected at '^' marker.
Router(config)#exit
Router#
*Jun 5 07:45:17.467: %SYS-5-CONFIG-I: Configured from console by console
Router#
Router#show
*Jun 5 07:45:23.731: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/3 from LOADING to FULL, Loading Done
Type "show ?" for a list of subcommands
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/2, 00:00:27 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:00:41 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    10.10.10.2, from 2.2.2.2, 00:00:27 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 59: config huawei pengaktifan

```

VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.40.1    4.052 ms  3.986 ms  4.814 ms
 2  10.10.11.2     7.592 ms  4.202 ms  4.325 ms
 3  *192.168.50.10  9.482 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS>

```

Gambar 60: trace 50.10 kembali aktif

2 Analisis Hasil Percobaan

2.1 Konfigurasi Cisco Switch

Pada praktikum pertama, kelompok saya melakukan konfigurasi VLAN dan trunk pada Cisco Switch. Hasil yang didapat menunjukkan bahwa VLAN 10, 20, 30, dan 40 berhasil dibuat dan aktif. Port access untuk masing-masing PC sudah sesuai dengan VLANnya, dan port trunk ke Cisco Router berhasil membawa keempat VLAN. Menurut teori, switch layer 2 membutuhkan VLAN untuk memisahkan lalu lintas jaringan. Kelompok saya berhasil menerapkan konsep tersebut. Yang mempengaruhi keberhasilan adalah ketelitian dalam memilih interface yang benar karena topologi memiliki

banyak port. Kelompok saya tidak menemukan kesalahan berarti, tetapi potensi kesalahan seperti salah memasukkan nomor VLAN atau lupa menyimpan konfigurasi dengan write memory bisa saja terjadi. Praktikum ini dinyatakan berhasil karena semua verifikasi dengan show vlan brief dan show interfaces trunk menunjukkan hasil yang sesuai.

2.2 Konfigurasi Cisco Router sebagai Router-on-a-Stick dan DHCP Server

Pada praktikum kedua, kelompok saya mengkonfigurasi Cisco Router sebagai gateway antar VLAN dengan metode router on a stick. Subinterface untuk VLAN 10, 20, 30, dan 40 dibuat pada interface Gi0/2 yang terhubung ke switch. Setiap subinterface diberi alamat IP yang menjadi gateway untuk masing-masing VLAN. DHCP server dikonfigurasi untuk VLAN 10 dan 20, sedangkan VLAN 30 dan 40 menggunakan IP statis. Hasil verifikasi menunjukkan subinterface memiliki IP yang benar, DHCP pool terbentuk, dan client VLAN 10 dan 20 berhasil mendapat IP dari range yang ditentukan. Pengujian ping dari setiap PC ke gateway masing-masing berhasil, dan ping antar VLAN juga berhasil karena router menghubungkan antar subnet. Teori menyebutkan bahwa router memungkinkan komunikasi antar VLAN yang berbeda subnet, dan kelompok saya berhasil membuktikannya. Ketelitian dalam menentukan encapsulation dot1Q dan nomor VLAN pada subinterface sangat berpengaruh.

2.3 Konfigurasi Huawei Router dan IP Address Antar Router

Praktikum ketiga berfokus pada konfigurasi alamat IP pada tiga router berbeda vendor (Cisco, Huawei, MikroTik) dan pengujian konektivitas langsung. Kelompok saya mengkonfigurasi LAN Huawei dengan network 192.168.50.0/24, kemudian memasang alamat IP pada empat link: Cisco-Huawei link 1 (10.10.10.0/30), Cisco-Huawei link 2 (10.10.11.0/30), Huawei-MikroTik (10.10.20.0/30), dan Cisco-MikroTik (10.10.30.0/30). Setelah konfigurasi, verifikasi menunjukkan semua interface dalam status up dan setiap router berhasil melakukan ping ke router tetangga langsung. Teori dasar routing menyatakan bahwa dua router dapat berkomunikasi jika berada dalam subnet yang sama dan memiliki IP yang benar. Kelompok saya berhasil membangun koneksi point-to-point pada semua link. Tantangan yang dihadapi adalah perbedaan sintaks perintah antar vendor, misalnya undo shutdown pada Huawei vs no shutdown pada Cisco, serta kebutuhan melakukan commit pada Huawei agar konfigurasi tersimpan.

2.4 Konfigurasi OSPF Multivendor dan OSPF Cost

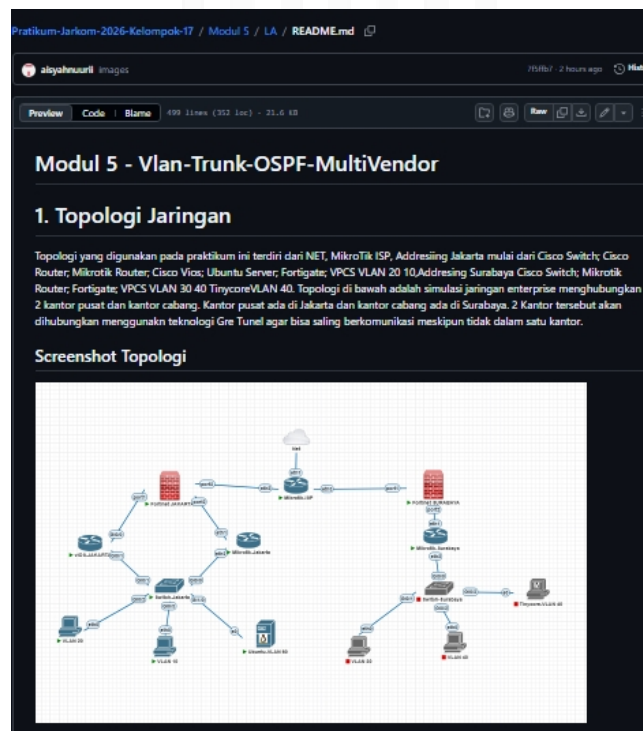
Pada praktikum keempat, kelompok saya mengimplementasikan routing dinamis OSPF pada tiga router berbeda vendor. Tujuannya adalah agar setiap router mengetahui rute menuju network di sisi lain secara otomatis, dengan jalur utama melalui dua link Cisco-Huawei (cost 10) dan jalur backup melalui MikroTik (cost 100). Hasil konfigurasi menunjukkan bahwa semua router berhasil membentuk neighbor OSPF. Cisco memiliki dua neighbor (Huawei dan MikroTik), Huawei memiliki dua peer (Cisco dan MikroTik), dan MikroTik memiliki dua neighbor (Cisco dan Huawei). Routing table OSPF pada Cisco menampilkan rute ke LAN Huawei 192.168.50.0/24, pada Huawei menampilkan rute ke VLAN 10,20,30,40, dan pada MikroTik menampilkan semua rute. Pengujian end-to-end dari PC VLAN ke PC LAN Huawei berhasil dengan ping reply. Teori OSPF menyatakan bahwa router dengan cost lebih rendah akan memilih jalur terbaik. Dengan cost 10 pada link langsung dan cost 100 pada link melalui MikroTik, maka jalur utama adalah melalui Cisco-Huawei.

2.5 Pengujian Failover OSPF

Praktikum kelima menguji kemampuan failover OSPF ketika jalur utama mengalami gangguan. Pada kondisi normal dengan dua link Cisco-Huawei aktif, rute dari Cisco ke LAN Huawei melewati jalur langsung dengan cost 10. Trace route menunjukkan jalur langsung Cisco ke Huawei. Ketika satu link dimatikan, rute masih tetap melalui jalur langsung yang tersisa karena cost tetap lebih rendah daripada jalur backup. Hal ini sesuai teori OSPF yang tetap memilih jalur dengan cost terendah meskipun hanya satu link tersisa. Ketika kedua link Cisco-Huawei dimatikan, OSPF secara otomatis mengalihkan lalu lintas ke jalur backup melalui MikroTik dengan cost 100. Trace route menunjukkan jalur Cisco → MikroTik → Huawei. Setelah kedua link utama diaktifkan kembali, OSPF mengembalikan rute ke jalur utama karena costnya lebih rendah. Proses perpindahan jalur berlangsung dalam hitungan detik. Hasil ini membuktikan bahwa OSPF dapat menyediakan redundansi dan failover otomatis.

3 Hasil Tugas Modul

Hasil jawaban dari tugas modul berada di github folder LA pada readme.md kelompok 17



Gambar 61: tugas modul 5

4 Kesimpulan

Berdasarkan percobaan percobaan dari pratikum 1 hingga 5 pada modul 5 ini, kelompok saya berhasil mengkonfigurasi VLAN, trunk, router-on-a-stick, DHCP server, koneksi antar router dari tiga vendor berbeda, serta routing OSPF multivendor dengan pengaturan cost. Hasil yang diperoleh sesuai dengan teori, di mana OSPF mampu memilih jalur utama melalui link langsung Cisco-Huawei dan secara otomatis beralih ke jalur backup melalui MikroTik ketika semua jalur utama terputus, kemudian kembali ke jalur utama setelah dipulihkan. Pembelajaran penting yang didapat meliputi perbedaan

sintaks antar vendor, pentingnya ketelitian dalam konfigurasi, serta mekanisme failover OSPF. Seluruh tujuan praktikum tercapai dengan hasil yang memuaskan.

5 Lampiran

5.1 Dokumentasi saat praktikum



Gambar 62: foto pratikum modul 5

